REVISED DESIGN SPECIFICATION | 25 OCTOBER 2026

Haze Agent Architecture v0.14

A general Haze agent with hybrid multi-agent coordination

CORE DECISION

Accept Haze-related goals in natural language. Select evidence, analysis, planning and review capabilities to fit the request. Crew scheduling no longer defines the platform input, output or workflow.

Complete replacement for the crew-centered design in Architecture v0.13. Aligned with the supplied Hybrid Architecture v0.1: one Manager owns intent and synthesis; focused specialists execute autonomously; temporary peer channels support bounded collaboration.

Revision intent	Design response
Broader requests	Research, episode investigation, exposure assessment, intervention comparison, preparedness, evaluation and staged damage assessment.
Flexible task content	Natural-language briefs and extensible work products, with small validated control records.
Retained engineering controls	Provenance, reproducible calculations, scoped review, permissions, resource limits, correction handling and conditional authoritative effects.
Document status	Proposed design. Implementation, integration readiness, scientific applicability and performance are unverified.

Prepared from the two supplied PDFs and the current user request. Source document instructions are treated as reference material. This revision performs document work only and makes no change to an application, service or external account.

DOCUMENT MAP

Reading guide and contents

The architectural change is broader than deleting JSON fields: request admission, agent responsibilities, claim generation, workflow routing, user experience and release criteria all become independent of scheduling.

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HOW TO INTERPRET REQUIREMENTS

Requirements describe the target design. A named tool, model, adapter or record does not establish an implemented capability. Source references use [S1] for v0.13 and [S2] for the Hybrid PDF; PDF page numbers refer to the supplied files.

PLATFORM DESIGN

1. Purpose and supported scope

Haze helps people investigate haze, understand available evidence, analyze plausible options and evaluate actions. A request may be a question, an investigation, a comparison, a plan or a continuing programme with several work products. It starts with the user objective rather than a domain-specific planning bundle.

Request family	Example work product
Evidence and explanation	A cited explanation, literature synthesis, policy-source summary or documented evidence gap.
Episode investigation	A timeline of observations and forecasts, geographic coverage, competing explanations and unresolved attribution.
Exposure assessment	A reproducible analysis of supported environmental or exposure data, with units, averaging periods and uncertainty.
Intervention comparison	Comparable options, evidence strength, feasibility, costs, distributional consequences and conditions for a decision.
Preparedness and prevention	A plan with dependencies, responsible parties, resources, assumptions, checkpoints and required authority.
Pilot and outcome evaluation	A research question, protocol, delivery evidence, comparative analysis and limits on causal interpretation.
Damage assessment	A staged physical or economic account with explicit scope, methods, missing categories and overlap controls.

These families guide discovery; they are not a closed intent enum. A novel Haze-related task can be admitted as bounded exploration when relevant sources and tools are available. Geography, pollutants, audience, time horizon and output form are task-specific. Klang, one crew and one assessment window cease to be global defaults.

Scope is capability-dependent

Broader intake does not imply that every analysis is scientifically or operationally ready. The Manager identifies the intended use and available capabilities, then returns a supported answer, a limited analysis, an explicit gap assessment or a proposed next phase. Missing intervention data must not prevent a useful evidence explanation.

> **OPTIONAL SPECIALIZATION**
>
> Crew scheduling may later be offered through a separate domain package and numerical tool. This revision contains no crew scheduling contract format. Such a package must be explicitly selected and cannot constrain unrelated requests.

Basis: [S1] sections 1 and 20-21; [S2] sections 3, 5 and 13.

ARCHITECTURAL CHANGE

2. What replaces v0.13

v0.13 assumption	v0.14 replacement
Only plan_outdoor_work is admitted	Open natural-language objective, intended use and desired deliverables.
An accepted planning bundle is mandatory before work	Start with the available request and context; acquire missing evidence through bounded assignments.
All work passes through a scheduler	Choose direct response, a specialist, parallel work, deterministic computation or a staged programme.
WorkerShield only selects fixed facts for an explanation	WorkerShield formulates analyses, runs permitted calculations, compares options and produces supported plans.
Ten fixed fact types and a universal controlled renderer	Evidence-linked prose for general synthesis; validated quantitative outputs and exact reviewed material where required.
Worker/activity protection and disruption ranking are core gates	Load explicit constraints and evaluation criteria appropriate to the named decision.
Scheduling feasibility and a four-day slice define success	Evaluate task completion, evidence quality, constraint preservation, user usefulness and cost across Haze tasks.
A general peer protocol and broad topics are deferred globally	Enable bounded peer collaboration in the core; admit each domain capability by its own readiness and authority.

Controls retained and generalized

- Preserve exact evidence versions, source locations, access rules, data lineage, units and material uncertainty.
- Keep execution success, evidence support, scientific applicability, review and authoritative state separate.
- Keep a single World Model authority for accepted domain state; runtime enforcement cannot be delegated to prompt text.
- Preserve immutable artifacts, explicit corrections, dependency invalidation, bounded retries and honest provider provenance.

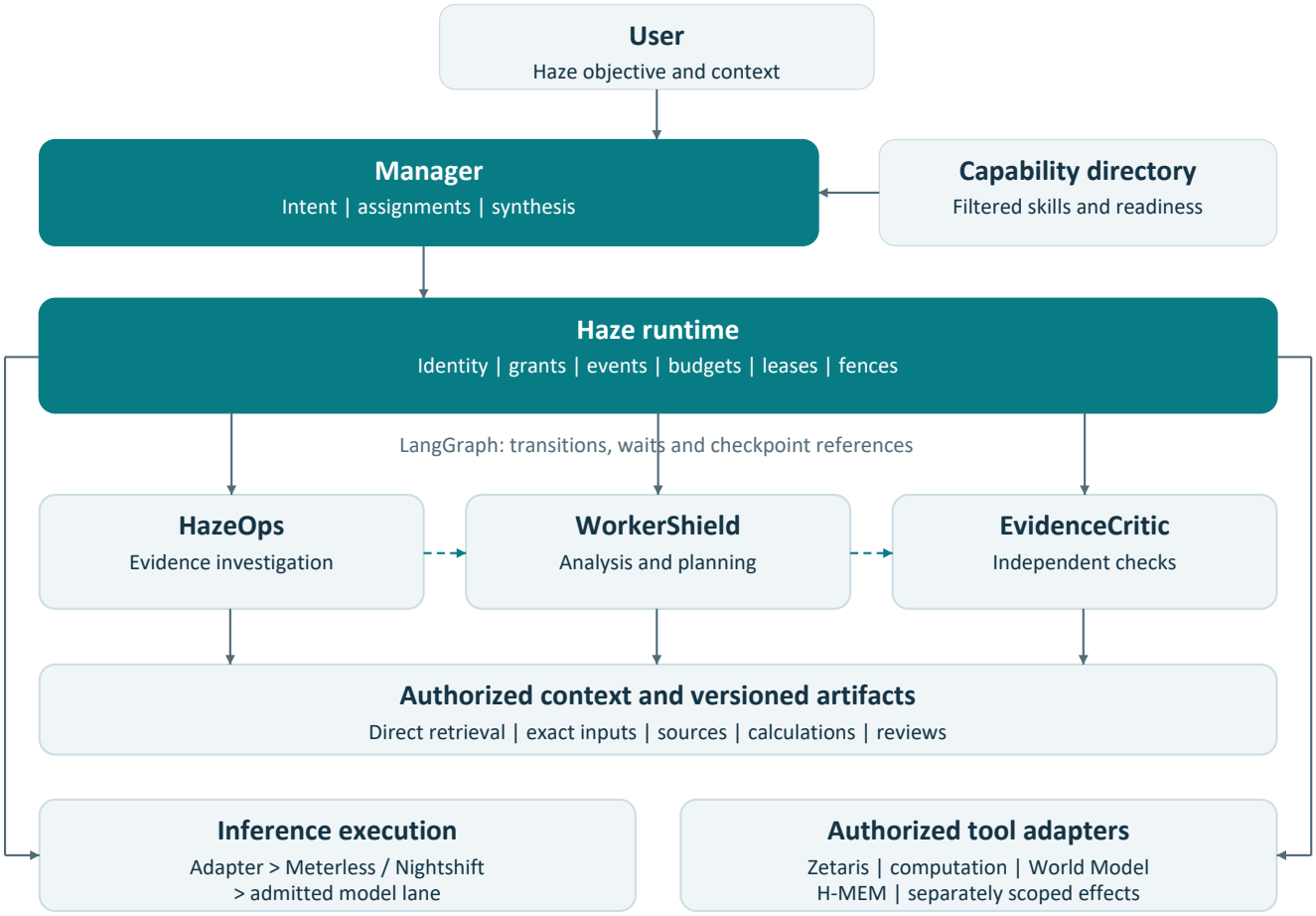
> **NO GLOBAL SCIENTIFIC APPROVAL GATE**
>
> General informational answers do not require operational approval or canonical publication. A consequential recommendation, accepted decision or external effect requires the checks applicable to that intended use.

Basis: [S1] sections 3, 5-9, 15-19 and 21; [S2] sections 3-5, 9 and 13.

HYBRID COORDINATION

3. Logical architecture

One application can host these logical boundaries. The Manager owns coordination; the runtime enforces control; agents own local execution; domain adapters supply evidence, computation and conditional effects. Separate services are an implementation choice after measurement.



Agents retrieve authorized versioned artifacts directly. The Manager inspects decisive evidence and objections without relaying every source byte or peer message. Inference execution is independent of permission to read data, invoke tools or commit domain state.

STATE OWNERSHIP

Coordination records own user intent and assignment state. World Model owns accepted domain decisions. The artifact store owns immutable working versions. LangGraph checkpoints reference these records; Zetaris and H-MEM do not become competing authorities.

Adapted from [S2] section 2, with the scheduling-first numerical path in [S1] section 2 removed.

PLATFORM DESIGN

4. Roles and local autonomy

Role	Responsibility and boundaries
Manager	Preserve the original objective, constraints and corrections. Choose work products, assignments and dependency versions; allocate resources; resolve conflicts; synthesize the answer. Cannot mint authorization or override review.
HazeOps	Investigate evidence and data. Retrieve originals, assess coverage and contradictions, prepare selections and lineage, and name gaps. Descriptive checks are allowed; scientific acceptance is not automatic.
WorkerShield	Frame analytical questions, formalize task-specific constraints, run reproducible computations, compare scenarios, assess feasibility and draft conditional plans. The name is retained; its scope extends across Haze analysis.
EvidenceCritic	Independently inspect sources, reproduce selected computations and assess assumptions, applicability and claim support. Record checks actually executed and unresolved objections. Cannot review its own producer work.
Runtime and LangGraph	Runtime admits and fences work, grants tools and resources, and persists events. LangGraph coordinates lifecycle transitions, waits and checkpoint references through those controls.
Human/domain owners	Supply operational facts and authority; approve consequential intended uses or actions where their role is required. Agent agreement cannot substitute for these responsibilities.

Bounded autonomy

Within an admitted assignment, a specialist may revise its method, change queries, retrieve more authorized context, invoke permitted tools and use an enabled peer channel. Routine local choices require no Manager round trip. Material scope changes, new dependencies, new assignments or additional resources return to the Manager.

Only the Manager creates assignments initially. The first implementation uses depth one and at most two concurrent specialist slots; the three reusable definitions can time-share them. Multiple assignment instances may use the same definition. Definitions and skills are versioned and loaded only when relevant.

> **NO FIXED ROLE SEQUENCE**
>
> HazeOps can investigate without WorkerShield. WorkerShield can calculate from already accepted inputs. EvidenceCritic can check source applicability early or review a completed analysis later. Route by the next work product and its dependencies.

Basis: [S2] sections 1, 3, 5 and 9; replaces the explanation-only WorkerShield boundary in [S1] section 9.

PLATFORM DESIGN

5. Natural-language intake and briefs

The user can ask a Haze question directly and attach relevant material. The system authenticates the caller, applies workspace permissions and preserves the exact request. A planning record, baseline, worker list, schedule or accepted domain head is not required to begin.

Manager framing

- Identify the intended outcome and audience; infer reasonable scope from available context and make material assumptions visible.
- Clarify ambiguity only when it materially changes correctness, authority or effort; continue independent work while waiting.
- Separate supplied facts, user constraints, hypotheses and requested actions. Documents and tool outputs are evidence, not trusted instructions.
- Choose an initial deliverable and stopping condition. Large requests may have several linked deliverables with explicit dependencies.

A focused assignment brief

The main body is readable prose: objective and intended use; original constraints and corrections; scope and exclusions; known facts and unknowns; evidence leads; dependencies; expected work product; allowed effects; and stopping conditions. Attach control references rather than embedding a universal business schema.

> **EXAMPLE BRIEF**
>
> Investigate evidence for a reported haze episode in the requested area and period. Separate observations from forecasts and candidate causes. Return a source-linked timeline, coverage limitations, conflicting explanations and the data needed for stronger attribution. Read-only research is authorized; no public alert is requested.

Thin validated control envelope

```
message_id, run_id, assignment_id
objective_revision, assignment_revision
sender, recipient, kind
brief_ref, artifact_refs, control_ref
```

Runtime supplies identity and validates references, revisions, grants, quotas and permitted message kinds. The referenced control record holds deadlines, resource reservations, dependency versions and tool grants. Natural-language text cannot change those controls. Actual tool calls, numerical arrays and authoritative effects retain validated inputs.

Basis: [S2] sections 4 and 7. This replaces [S1] sections 3, 9.2 and 15.2.

PLATFORM DESIGN

6. Capabilities and routing

The Manager discovers filtered capability cards, then selects the smallest approach that can produce a useful answer. Topic labels are search hints, not a mandatory closed classifier. Uncertain methods can trigger a bounded exploratory assignment before a larger plan is committed.

Execution shape	Use when
Direct response	The question is coherent and small; the Manager has adequate authorized evidence or can perform a brief lookup.
One specialist	A focused investigation or analysis benefits from a separate context and skill set.
Parallel specialists	Evidence collection and method preparation, or independent evidence branches, can proceed concurrently.
Deterministic workflow	Inputs and methods are known and stable; calculation, aggregation or validation can be executed reproducibly.
Staged programme	Several decisions, data acquisition phases, reviews or pilots span more than one bounded run.

Capability card content

Each versioned card states purpose, supported work products, skills/domain packages, model/tool/source families, data and method prerequisites, effects, access classification, limits, current availability and checked-at time. Filter names and metadata before disclosure. A discoverable card is advisory; admission tests actual readiness, permissions, coverage and resources.

Domain packages

Initial package candidates include episode reconstruction, environmental data quality, exposure analysis, intervention evidence, local guidance, preparedness, evaluation and damage accounting. A package supplies retrieval guidance, methods, data definitions, applicability limits and checks. It is not a permanent new agent or a new core task format.

A package that is absent, unsupported or unreviewed causes a scoped blocker. The Manager can still answer adjacent supported questions or specify the evidence needed. Adding a topic should require registering a capability and its tests, rather than changing the core request/result schema.

> **CONSTRAINTS FOLLOW THE TASK**
>
> Honor explicit constraints such as budget, geography, affected population or no increase in modeled burden. Do not inherit fixed assignments, timing grids, movement penalties or a universal percentage target from v0.13.

Basis: [S2] sections 3-5 and 12-13.

PLATFORM DESIGN

7. Assignment lifecycle

Step	Required behavior
Frame	Preserve user intent, intended use, scope, unknowns and the next decision. Choose an initial work product.
Plan	Create a small dependency graph. Pin required artifact versions; reject cycles. Admit independent work in parallel.
Admit	Persist assignment identity and definition version; reserve aggregate resources; bind a lease, grants, context and tools. Return a concrete admission/rejection reason.
Execute	Specialists select local methods, retrieve authorized sources and save substantial intermediate artifacts. Progress reports interpreted scope, evidence and blockers.
Coordinate	Use targeted steering and material questions. Enable a peer channel only for a concrete exchange; waits release execution slots.
Produce	Return the work product, exact dependencies, assumptions, uncertainty, coverage and unfinished checks for the objective revision used.
Check	Apply deterministic checks and independent review appropriate to the claim and intended use. Revise only the affected branch.
Synthesize	Manager inspects decisive artifacts and objections. Deliver a current supported answer or an explicit partial result with next requirements.

States and events

Execution states can include queued, running, waiting, paused, completed, failed, cancelled and timed_out. Output usability and evidence support are separate from execution state: a completed assignment can return an important gap, and a previously completed artifact can later become stale.

Use durable events for progress, questions, partial artifacts, failures, revision acknowledgement and results. Persist delivery cursors and deduplicate messages. Downstream work consumes explicitly usable artifact versions, not a generic completed flag. Provisional inputs support labeled exploration; they do not silently become accepted evidence.

> **SUCCESS IS TASK-SPECIFIC**
>
> An adequate explanation, a documented contradiction, a reproducible comparison or a justified finding that evidence is insufficient can each satisfy a request. The system does not have to generate a preferred intervention to count as useful.

Basis: [S2] sections 5 and 9. Replaces the planner-first path in [S1] section 16.

PLATFORM DESIGN

8. Context, artifacts and memory

Focused context with retrievable originals

Initialize each assignment with the relevant brief, original user corrections, current decisions, evidence leads and known omissions. Preserve the difference between trusted instructions, the user request, Manager interpretation and retrieved material. A retrieved document or peer message cannot install instructions or enlarge grants.

Retrieve authorized originals and neighboring passages as needed. Permission filtering applies before metadata, excerpts, caches or model-provider disclosure. Record search scope, filters, inaccessible material, truncation and gaps. A search with no result supports only a statement about that search, not a claim that evidence does not exist.

Small common artifact metadata

Always applicable	Additional metadata when relevant
Identity, type and immutable version/hash	Original source location, raw bytes/hash and retrieval receipt.
Producer and assignment/objective revision	Query, selected rows/intervals, transformations and method versions.
Scope, status, access classification and dependencies	Units, spatial/temporal support, uncertainty and intended-use limits.
Created-at time and supersession links	Calculation inputs, code, parameters and environment for reproduction.
Permission and retention controls	Reviewer identity, executed checks, applicability, expiry and recheck conditions.

Artifacts can be prose, evidence maps, tables, datasets, notebooks, calculations, plans, reviews or approved action payloads. Each type validates its own relevant content. Optional extensions must not become a giant record with mandatory fields for every Haze domain. Large outputs are referenced rather than copied through all contexts.

Compaction and reuse

Working notes retain constraints, hypotheses, contradictory evidence, failed searches, open questions and next steps. Compaction includes original references and known omissions. Inspect decisive originals before consequential synthesis. Summaries do not establish complete source coverage.

H-MEM stores eligible reviewed reusable lessons with support, scope and expiry. It is neither the live scratchpad nor the current domain authority. Revalidate access and support before reuse; report recall or capture failure independently. Memory must not contain held-out evaluation answers or silently promote speculative learning.

> **ACYCLIC PROVENANCE**
>
> Inputs precede producer artifacts; independent checks reference those immutable artifacts; reviews reference completed material. Later checks never require rewriting a producer hash to insert references to their own future records.

Basis: [S1] sections 4, 10, 13 and 17; [S2] section 7.

PLATFORM DESIGN

9. Evidence, claims and review

General research needs supported synthesis in normal language. The fixed ten-type fact vocabulary and ID-only explanation plan from v0.13 are retired as universal output requirements. Flexible prose still carries evidence, scope and uncertainty; technical validation remains mandatory at calculation and effect boundaries.

Material type	Support and checking
Source-backed description	Cite the exact document passage or dataset selection, source kind, date and relevant scope. Separate the source assertion from Haze interpretation.
Synthesis or inference	Link supporting and conflicting evidence; explain the inference and its assumptions. Do not present inference as a direct observation.
Quantitative result	Pin inputs, method/code, units, transformations, numerical checks and reproduction instructions. Generate displayed values from checked artifacts.
Recommendation or plan	State decision criteria, alternatives, assumptions, uncertainty, affected groups, feasibility and required intended-use review.
Accepted decision, alert or action payload	Bind the exact material content to review, authority, current dependencies and a conditional commit/effect receipt.

Review depth follows the task

For a low-consequence factual lookup, source inspection and citation validation may suffice. A broad investigation may commission EvidenceCritic on decisive claims or conflicting evidence. Consequential operational analysis requires independent numerical/constraint checks and the qualified domain review appropriate to the intended use. The Manager records why the review scope is adequate.

An independent reviewer uses authorized originals or reproducible checks rather than producer reassurance. Record reviewer identity, separation from producer, exact artifact versions, checks performed, findings, unresolved objections and limits. Model agreement, schema validation and hashes are not evidence of truth or scientific applicability.

Exact material where authority requires it

Before an accepted decision or consequential external payload is approved, finish material wording, numbers, caveats, provenance and hashing. Approval binds that exact version. A material edit requires a new version and affected re-review; presentation-only formatting may preserve approval if content is unchanged. Ordinary conversational synthesis does not need a universal controlled sentence template.

> **TRUTHFUL PARTIAL OUTPUT**
>
> An unresolved numerical claim can be withheld while supported findings and the precise gap are delivered. Missing values remain missing; failed or pending checks cannot support a positive assertion.

Basis: generalized from [S1] sections 4, 8 and 11; aligned with [S2] sections 3, 5 and 7.

PLATFORM DESIGN

10. Domain methods and applicability

Use deterministic tools for calculations and domain packages for methods and applicability. The core coordinates work; it does not prescribe one exposure proxy, optimization objective or scientific model for every request. Each activated method declares scope, required inputs, assumptions, limitations, version and validation evidence.

Boundary	Required distinction
Environmental evidence	Separate observations, forecasts, modeled fields and synthetic examples. Preserve event time, issue time, valid interval and retrieval time. Re-fetching does not make old evidence new.
Quantity and resolution	Preserve pollutant, unit, averaging period, native resolution and spatial support. A finer computation grid does not create finer environmental information.
Exposure analysis	Require supported time/location/population inputs for the claim being made. Concentration-time arithmetic does not by itself establish dose, safe duration or health benefit.
Intervention comparison	Use an explicit comparator and comparable endpoints. Do not invent filtration ratios, protection factors, implementation effectiveness or adoption.
Attribution	Hotspots, smoke signals or temporal correlation alone do not establish source responsibility, downwind contribution or avoidable damage.
Policy and health sources	Use the relevant original authority and revision. A summary cannot grant legal authority, clinical suitability or scientific approval.

Uncertainty and negative findings

Separate data gaps, parameter uncertainty, structural alternatives and analyst scenarios. Keep correlations and paired comparisons where required by the method. A sensitivity is not a probability interval unless its probabilistic basis is justified. A supported result for one population, place or period does not establish transferability to another.

State the exact searched or modeled scope. An incomplete search cannot prove absence, global optimality or target unattainability. A checked witness can establish a scoped possibility without proving that it is the best option. Search and optimization fields appear only for tasks that perform those operations.

> **METHOD READINESS IS USE-SPECIFIC**
>
> Research exploration may continue with explicit assumptions where authorized. Operational exposure guidance, epidemiological inference, economic valuation and causal evaluation require their own suitable evidence and qualified review. No parameter or numerical threshold is newly adopted by this revision.

Basis: [S1] sections 4-7 and 20-21; [S2] sections 3, 7 and 10-12.

PLATFORM DESIGN

11. Corrections and invalidation

A user correction changes the coordination record immediately. Preserve the exact original passage, advance the objective revision and identify affected assignments and artifacts through pinned dependencies. When impact is uncertain, mark the affected branch stale conservatively.

Control point	Required result
Persist correction	Record the request and revised constraints durably before acknowledging receipt. Distinguish receipt from operational activation.
Find dependents	Invalidate affected briefs, calculations, rankings, claims, reviews and unsent actions transitively. A completed artifact can become stale.
Steer workers	Send the targeted delta and current brief. Require revision acknowledgement at a safe boundary before further affected tool admission.
Handle peer grants	Close channels bound to affected revisions. Re-admit a still-useful exchange with current participants and limits.
Reuse selectively	Retain unaffected snapshots or calculations only after an explicit recorded impact/applicability check. Do not relabel an old result as new.
Fence effects	Block affected effects immediately; synchronize a canonical hold/generation before claiming the correction is operationally active. Reconcile in-flight receipts.
Complete	Recompute and re-review affected work or deliver a partial response naming stale/blocked branches. Preserve budgets and deadlines.

Two revision domains

The objective revision versions user intent and constraints. Accepted domain/support revisions version evidence, methods, decisions and holds within World Model. A conversational correction does not silently accept a new scientific fact. An action-affecting correction must also update the relevant authoritative fence before new effects can proceed.

Late messages remain associated with their original assignment revision and lease. The runtime excludes them from current synthesis and commits unless explicitly revalidated. Delivery, acknowledgement and dependency reconsideration are distinct events; a message receipt is not proof that stale work has stopped.

> **EXAMPLE**
>
> If the user removes an intervention from the feasible scope, invalidate that option and all dependent comparisons, recommendations and reviews. Preserve independent episode observations after checking their continued relevance. Reassess alternatives under the new constraint rather than changing only the final paragraph.

Basis: [S2] section 8 and [S1] sections 10-13.

PLATFORM DESIGN

12. Runtime controls and long requests

Broad scope is implemented through bounded work, durable state and explicit continuation. The programme may persist across runs; an individual model/tool execution remains finite. The old universal 300-second scheduling pipeline and its fixed search-stage allocations are retired.

Runtime control	Enforced behavior
Permissions	Intersect user/workspace policy, definition and assignment grants. Check every tool, artifact read, metadata disclosure and effect. Text cannot escalate permissions.
Aggregate budget	Reserve time, tokens/spend, tool attempts, artifacts and peer traffic across Manager, specialists and review. Preserve synthesis/review capacity; no multiplied hidden retries.
Admission profile	Choose limits appropriate to a short lookup, investigation or staged study. Persist absolute deadlines and stopping rules; numeric defaults remain configurable, measured engineering choices.
Durability	Persist assignments, revisions, events/cursors, operation keys, leases and checkpoints. Assume repeated delivery and deduplicate.
Recovery	Reload current grants, dependencies and remaining reservations under a new lease generation. Fence the old worker. Resume preserves charged attempts and the original deadline.
Pause/cancel/revoke	Stop new affected admission and close channels; cooperatively interrupt or fence in-flight work. Report requested versus confirmed control state.
Retries and effects	Use one bounded attempt ledger. Reconcile an unknown effect outcome with the receiving service before retrying; record possible charges.
Partial completion	Retain usable versioned findings and name missing work. Do not label a partial answer complete or claim remote computation/billing stopped without evidence.

Larger studies

Split a long request into coherent deliverables and phases with a shared objective and dependency graph. Each new run is linked, explicitly admitted and independently bounded; it does not inherit permission to spend without limit. Human/data waits persist as waiting state and release execution capacity. Continuation after expiry receives a new reservation and current dependency check.

> **INITIAL CONCURRENCY**
>
> Start with two specialist execution slots, Manager-only assignment creation and depth one. Increase concurrency only after measuring cost, coordination overhead, quality and latency. A durable programme is not an endless model loop.

Basis: [S1] section 13 and [S2] sections 5 and 9.

PLATFORM DESIGN

13. Temporary peer collaboration

Use peers for specific evidence clarification or reproduction questions. The Manager opens or widens a channel subject to runtime admission. Specialists may request one or use an already granted channel without approval for each message. There is no default fully connected mesh or peer spawning.

Grant property	Rule
Participants	Named active assignment instances and revisions; a reusable role name is insufficient.
Purpose	A concrete question such as a timestamp definition, a source passage or a calculation input.
Permitted content	Natural-language clarification, authorized excerpts, artifact references and inputs within both the purpose and each recipient's access.
Bounds	Explicit expiry, message/byte limits and relevant tool/spend limits charged to the same parent run. No automatic renewal.
Visibility	Durable messages and references available to the Manager and authorized audit readers. Surface material changes without relaying routine messages.
Closure	Close on completion, expiry, exhaustion, revocation, cancellation or relevant revision change. Persist a short outcome and unresolved issues.

ILLUSTRATIVE EXCHANGE

HazeOps and WorkerShield may reconcile whether a data timestamp denotes interval start or end. A proposed test grant allows at most six messages and 12 KB for ten minutes, shortened by the run deadline. These are example test limits, not domain rules.

Runtime filters metadata, snippets and artifact dereferences for each recipient before disclosure. An artifact identifier does not confer access. A redacted derivative may be shared if it preserves meaning and is authorized; otherwise return a blocker.

Peers cannot alter the objective, create assignments, widen grants, allocate budget, authorize external effects or promote a conclusion into accepted state. Escalate unresolved disagreement, scope change or missing resources to the Manager. EvidenceCritic may request reproduction instructions, then inspect originals or rerun checks independently; producer clarification is not verification.

Basis: [S2] section 6. This replaces the general deferral of peer collaboration in [S1] section 21.2.

PLATFORM DESIGN

14. Integration boundaries

Boundary	Responsibility and readiness evidence
Haze runtime / LangGraph	Runtime owns identity, admission, quotas, grants, channels, leases and fences. LangGraph drives checkpointed lifecycle transitions. Verify duplicates, restart, cancellation and fencing in the actual implementation.
Inference adapter / Meterless Runtime / Nightshift	Execute admitted Manager/specialist model turns. Verify reachability, authentication, generation, tool/context behavior and actual model route. Model execution supplies no data/effect authority.
Zetaris	Authorized retrieval/query boundary. Record view/query versions, selected rows/time/locations, upstream revisions or hashes, transformations, missingness and coverage. A reachable view may be insufficient.
World Model	Sole authority for accepted/current domain state, decisions and holds. Verify exact conditional writes, durable events and current-read fences in the installed backend.
H-MEM	Reviewed reusable learning with provenance, scope and expiry. Revalidate permissions and support on recall; capture failure/lag remains separate from result validity.
Context/artifact adapters	Preserve original user passages and immutable evidence, calculations and reviews; enforce access filtering and retrieval of exact versions.
Computation/domain adapters	Run reproducible data preparation, numerical analysis or planning with validated inputs. Register actual methods and readiness rather than adding a universal planner to the core.

Provider provenance and degradation

Keep the selected Meterless Runtime/Nightshift architecture from the supplied sources. A WorkerShield Nemotron feature is attributed only when its actual feature-correlated receipt/log identifies that model lane. Catalogue entries and unrelated generation do not prove feature execution. Other agent bindings need their own evidence.

Admit an explicit provider/failure policy. Do not silently switch routes, providers or live/sample modes. If a declared fallback uses checked deterministic output, label it accurately and preserve failed inference evidence. Fallback cannot invent an investigation or waive required review. Store no secrets in artifacts, logs, browser assets or public files.

> **DESIGN STATUS**
>
> These are logical responsibilities and verification requirements, not claims about installed vendor APIs or working integrations. No inherited hackathon or sponsor requirement is adopted as a current rule without separate verification.

Basis: [S1] sections 2, 14 and 17; [S2] section 10.

PLATFORM DESIGN

15. Accepted state and external effects

Three distinct output levels

Level	Authority
Working output	Notes, sources and exploratory analysis in immutable artifacts. May be useful but provisional, partial or unreviewed.
Reviewed conclusion	An exact artifact with named checks, findings and limits. It may remain scientifically uncertain or unsuitable for an operational use.
Accepted/current domain decision	An explicitly accepted World Model record with current support, exact review and applicable permissions/holds. A reviewed conclusion is not automatically accepted.

The Manager owns coordination, not canonical domain truth. General informational answers can be delivered as source-backed responses without canonical promotion. If the authoritative boundary is unavailable or unproved, authorized research can continue; accepted/current claims and affected effects remain blocked. Clearly label historical or exploratory material.

Conditional authoritative operations

The adapter must provide the logical ability to read a pinned snapshot, propose exact material, record independent review, conditionally accept/publish against current support and read current state. These are required semantics, not asserted vendor method names. Atomically check relevant generations, dependencies, holds, access, review/source validity and the effect authorization; append the committed record and durable event together where required.

Acceptance/hold and publication serialize on the same relevant authority. A correction between review and commit must fail the stale commit even if event consumers are paused. Current reads recheck relevant support and expiry rather than trusting cached flags. Unrelated changes should not invalidate an unaffected branch; ambiguous relevance creates a scoped hold or routing barrier.

Effects are separately scoped

Sending an alert, contacting a person, purchasing equipment, changing an operational system, dispatching a field team or releasing funds needs the appropriate existing user/operator authority and a ready connector. Exact target/content, review, operation key, expiry and receipts are retained. A plan or analysis does not grant action permission.

> **UNKNOWN COMMIT OUTCOME**
>
> After a timeout, inspect the receiving service using the stable operation key before retrying. Record idempotency/conditional-update support and reconcile in-flight effects on cancellation. A checkpoint, peer message or completed assignment is not an effect receipt.

Basis: [S1] sections 10-12 and 21.3; [S2] sections 7-10.

PROPOSED INTERFACES

16. API and result design

Introduce a new general interface rather than reinterpreting the legacy scheduling operation. All names below are proposed application interfaces and require implementation. Keep prose flexible while validating identity, revisions, refs, size limits, tool inputs and effect controls.

Interface	Purpose
POST /tasks	Authenticate and admit an objective plus optional context references, scope, constraints and desired deliverables. An accepted domain bundle is optional.
GET /tasks/{id}	Read authorized state, work products, gaps and review/currency.
POST /tasks/{id}/updates	Persist a correction or clarification against the expected objective revision; return receipt and affected-work status.
POST /tasks/{id}/control	Request pause, cancel or resume; return confirmed state and unresolved in-flight operations.
GET /tasks/{id}/events	Read progress, questions, results and acknowledgements after a cursor.
GET /artifacts/{id}	Read an authorized immutable version or exact source location. Filter metadata before disclosure.
GET /capabilities	Return filtered capability, version, prerequisite and readiness metadata.

Minimum result envelope

```
task/run identity + objective revision + execution state
answer/work products + completion/coverage scope
sources + applicable calculation/check references
assumptions + uncertainty + unresolved questions
review scope/status + as-of/currency + next requirements
actual tool/model provenance
```

A result need not contain a selected plan, baseline, worker score or optimizer certificate. Attach relevant validated domain artifacts. Answers can be prose, comparisons, tables, reports or plans; generic metadata stays stable as topics expand.

Scope idempotency keys to principal, workspace, operation and interface version plus an exact fingerprint. Same key/content returns the original result; conflicting reuse rejects. Distinguish malformed input, denied access, stale revision, unsupported capability and unavailable required adapters. Insufficient evidence for a valid request is a domain outcome rather than a server fault.

> **LEGACY RULE**
>
> Archive the old /run and haze.run.v2 semantics unchanged. Reject legacy scheduling payloads on the new general endpoint with explicit migration guidance. Existing approvals do not transfer to revised artifacts.

Replaces [S1] section 15; aligns with the small coordination interface in [S2] section 4.

PLATFORM DESIGN

17. User experience and monitoring

A general task workspace

Start with the user objective and attached context. Show the interpreted scope, current assumptions, requested deliverables and any material question. Provide progress by work product, an evidence drawer, analysis artifacts, review findings and useful next requirements. The default screen is no longer a supervisor schedule dashboard.

Visible information	Why it matters
What Haze is answering	The original request, current scope and accepted corrections remain visible.
What was found	Supported findings and relevant conflicting evidence lead the answer.
What is calculated or inferred	Distinguish observations, forecasts, computations, scenarios and interpretation.
What is missing	Name exact coverage gaps, unavailable sources and unchecked assumptions with their consequences.
What is reviewed/current	Show review scope and as-of status; do not use one badge to imply universal validation.
What happens next	Offer the next evidence need, analysis phase or separately authorized action with concrete conditions.

Request-driven monitoring

Monitoring is a capability activated by a user or accountable operator for a named source, topic, area, period and purpose. Define access, cadence, quotas, cursor, catch-up limit, deduplication, next due check and notification conditions. A manually executed lookup is not continuous monitoring.

Each scheduled tick is a bounded job. Detect revisions, retractions, duplicates and late arrivals; route relevant changes through exact dependencies and scoped subscriptions. Persist last attempt/success and source/routing lag. A source outage, absent detection or missing category never means all-clear or zero harm.

Notify only under the chosen policy and meaningful changes, completion, failure or required user action. Draft/public alert generation and sending are separate capabilities; recipient rights, language, place/time scope, exact content, expiry, review and delivery/correction receipts must be ready before external delivery.

Modes and honest degradation

Keep live, captured/replay and offline-sample execution distinct from data origin and authority level. Live inference on a synthetic example does not make the evidence real. Offline sample mode performs no remote calls and labels skipped integrations. Replay uses the original information cutoff and cannot claim a current environmental assessment.

Basis: [S1] sections 12, 16-17 and 21; [S2] sections 7, 9 and 10.

ILLUSTRATIVE WORKFLOW | NO LIVE DATA

18. Episode investigation trace

> **USER REQUEST**
>
> Investigate the reported haze episode in the selected area last week. Explain what the evidence supports, which causes remain uncertain and what data would improve the conclusion.

Stage	Expected behavior and work product
Frame	Manager resolves the requested place and exact dates, preserves the request and distinguishes episode description from source attribution. No schedule or operational baseline is requested.
Discover	Check available environmental, meteorological, satellite, official-report and contextual sources with appropriate access and temporal/spatial scope.
Parallel work	HazeOps builds a source-linked timeline and coverage map. WorkerShield prepares a method for comparing time/space patterns and assessing alternative explanations.
Clarify	A temporary channel can reconcile timestamp or averaging definitions. Artifact access is filtered independently for each assignment.
Analyze	Only supported comparable data enter the calculation. Record selections, units, missing intervals and method/code versions. Treat candidate causes as hypotheses unless an adequate attribution method supports more.
Review	EvidenceCritic inspects decisive sources and checks selected transformations or calculations; records contrary evidence, gaps and claim limits.
Synthesize	Manager answers with the episode timeline, supported patterns, competing explanations, confidence/uncertainty basis and exact next evidence needs.

A partial evidence branch

If satellite retrieval fails but authorized surface observations are usable, deliver the supported observation timeline with the missing satellite branch visible. Do not infer a source from the absent data. If hourly source data contain gaps, retain them and narrow any quantitative claim to supported intervals rather than interpolating silently.

Correction example

If the user corrects the area boundary, advance the objective revision, invalidate area-dependent selections, calculations and review, and reuse unaffected source definitions only after an impact check. Late old-area results remain historical. The final response identifies the current scope and any unresolved branch.

This trace specifies coordination behavior. It reports no retrieved episode data, successful model execution, validated attribution or measured runtime benefit. It replaces the two-worker fixture as a core demonstration case.

Basis: [S2] sections 5-9 and 12-13; broader programme scope from [S1] section 20.

ILLUSTRATIVE WORKFLOW | NO EFFECT ESTIMATES

19. Intervention comparison trace

USER REQUEST

Compare portable filtration, a cleaner-air room and a public information campaign for the selected community. Respect the budget, identify feasibility gaps and propose a pilot before recommending wider use.

Coherent parallel assignments

The Manager records the audience, place, horizon, budget and desired comparison. HazeOps investigates evidence, local facilities and source quality. WorkerShield frames comparable decision criteria, data needs, feasibility and pilot options. These assignments can overlap without calculating missing inputs as known.

An evidence table separates published intervention findings from local applicability. A plan table separates available facilities from verified usable capacity, equipment specification from delivered performance, and communication reach from changed exposure or health outcomes. Without a common supported endpoint, use a transparent multi-criteria comparison rather than inventing a single effectiveness score.

A scoped exchange and review

A peer channel can clarify whether an equipment figure is a specification or a measured result for the local room. WorkerShield pins the answer and source version. EvidenceCritic independently checks the decisive source passage, any numerical method and the scope of claimed effectiveness. Producer clarification remains labeled as clarification.

User correction during work

The user says: exclude public messaging and do not depend on a new building. Manager preserves that correction and advances the objective revision. Runtime fences the excluded branches and invalidates affected comparison, budget and draft-plan material plus its review. Existing room measurements and equipment evidence may be reusable after an explicit impact check.

Useful bounded outcome

The answer compares the remaining feasible options, states unresolved room/operation/maintenance inputs, and proposes a pilot protocol with outcomes, measurements, implementation checks, decision criteria and review requirements. Missing local evidence may prevent a supported ranking while still allowing a useful evidence review and data-collection plan.

AUTHORITY AND FAILURE

No purchase, public communication, facility activation or field deployment follows from this request. If the model route fails after an artifact is saved, preserve the current artifact, record the failed/unknown route and recover only under the admitted policy and remaining budget. A declared deterministic fallback cannot claim the agent investigation succeeded.

This is a design trace without supplied local data, estimated effect sizes or scientific approval. It demonstrates larger scope without scheduling-specific inputs. Basis: [S2] sections 8-11 and [S1] sections 20-21.

ENDURING HAZE PROGRAMME

20. Damage, prevention and evaluation

Retain the broad programme in v0.13 while removing the requirement to enter through crew scheduling. Damage assessment, prevention investigation and pilot evaluation become separate capability families that share evidence, analysis and review infrastructure.

Staged damage assessment

- First define pollutant/source scope, geography, population, period, endpoints, comparator/counterfactual and intended decision.
- Separate environmental concentration, population exposure, health outcomes and economic quantities. An air-quality indicator is not automatically a damage estimate.
- Activate attribution, health-response or valuation methods only with suitable inputs, reproducible code and qualified intended-use review. Preserve nonlinearities, incidence conventions, spatial/temporal coverage and transfer limits.
- Keep physical outcomes, financial incidence and selected social-welfare accounts distinct. Record overlap groups, excluded duplicates, unpriced effects, currency/price year and discount assumptions where relevant. Missing damage is not zero.
- Distinguish historical source-removal estimates from achievable benefit of a specific intervention, which depends on delivery, uptake, maintenance, persistence and displacement.

Prevention and accountable delivery

Investigations may examine source-prevention mechanisms or community preparedness where evidence and practical authority exist. Detection alone does not establish responsibility, preventability or benefit. Before an action, identify the operator, land/facility rights, resources, permissions, exact reviewed proposal, implementation record and independent delivery verification.

Pilot and learning lifecycle

Use hypothesis -> prespecified protocol -> design review -> authorized simulation/pilot -> implementation and deviation records -> observations -> comparative estimate -> independent result review -> accepted learning. Distinguish planned delivery, reported completion, verified completion and evaluated effect. A before/after change alone does not establish causal benefit.

Experiments specify outcome, population, comparison, measurements, analysis, uncertainty, missingness, stopping criteria and approvals. Do not invent sample size or health-effect parameters from a demonstration. H-MEM receives eligible reviewed lessons; World Model retains accepted decisions and evaluation provenance.

> **PROGRAMME TARGET**
>
> The earlier 10% damage-reduction ambition may remain an owner-adopted target after its metric, scope, comparator and evaluation criterion are defined. It is not a universal response gate, intervention benefit floor or result of this architecture revision.

Basis: [S1] sections 1, 18.3 and 20-22; [S2] sections 3 and 13. No scientific or monetary parameter is adopted here.

PLATFORM DESIGN

21. Evaluation and release gates

Freeze representative clean and corrupted cases before tuning. Include a short lookup, episode investigation, evidence synthesis, intervention comparison, preparedness plan, pilot evaluation and missing-evidence case. Test new topics without modifying the core request/result model. Separate development and held-out evidence, and preserve decision-time cutoffs.

Evaluation area	Required evidence
Evidence and claims	Source/citation accuracy, lineage, contradictions, coverage and applicable caveats; unsupported material claims and false abstentions by case.
Analytical quality	Independent reproduction of numerical outputs; units, time alignment, missingness, assumptions, sensitivity and limits on causal/health claims.
Intent and correction	Original/corrected constraints preserved; affected dependents found; stale results excluded; unaffected work reused only after impact checks.
Authority and recovery	Denied metadata/artifact access, injected document instructions, self-review, revoked permissions, duplicate operations, expired leases, cancellation and unknown commit outcomes.
Hybrid value	Compare with one capable agent using the same sources/tools and with a deterministic workflow where applicable. Add a hybrid-without-peer-channels comparison.
Efficiency and usefulness	Completion quality, human task usefulness, aggregate cost including review/retries, p50/p95 latency, coordination overhead and duplicated work. Report sample sizes and uncertain charges.

Capability-specific release

For the enabled task class, require captured end-to-end execution, adequate evidence/claim quality, suitable method review and passing runtime invariants. Test stale/unauthorized effects against the actual authoritative adapter; a local mock cannot establish its guarantees. Any observed unauthorized effect or stale accepted commit blocks the affected capability release.

For simple answers, verify useful completion without unnecessary agent orchestration. For complex work, compare quality at matched budgets as well as cost at comparable quality. A second agent is justified by measured benefit, not by the architecture diagram. Partial answers must be compared as partial answers.

> **UNEXECUTED APPLICATION CHECKS**
>
> This revision validates document structure, source alignment and PDF layout. It does not execute Haze application tests, provider calls, authority races, user studies, field experiments or scientific validation. All release gates above remain implementation requirements.

Basis: [S1] section 18 and [S2] section 12.

PHASED DELIVERY | NO ASSUMED EVENT DEADLINE

22. Implementation sequence

Replace the old four-day scheduling plan with dependency-based phases. Scope and staffing determine duration; the current user request supplies no delivery deadline. Retain one application and the existing boundaries while implementing one complete general path first.

Phase	Work and exit evidence
A. Remove core coupling	Inventory code/configuration for scheduler-only assumptions. Register general task/control/artifact interfaces, request framing, capability discovery and source-backed output. Exit: a non-scheduling request reaches a useful answer without a planning bundle.
B. Deliver a hybrid vertical path	Admit HazeOps and WorkerShield, persist events/artifacts, enable one bounded channel, independently check a decisive artifact and synthesize. Exit: a genuine trace with actual tool/model provenance and explicit gaps.
C. Make correction and recovery work	Exercise user steering, branch invalidation, revision acknowledgement, late results, duplicate delivery, cancellation, restart and permission revocation. Exit: captured invariant checks and unchanged remaining-budget semantics.
D. Broaden domain capability	Add evidence-backed domain packages and deterministic methods by task class; test cross-topic requests and intended-use review. Exit: held-out useful completion across the selected Haze cases.
E. Enable accepted decisions/effects selectively	Verify actual World Model current-state/commit fences and each effect connector; add exact material review, operation receipts and correction handling. Exit: capability-specific race/recovery/authority evidence.
F. Evaluate and operate	Benchmark alternatives, tune resource profiles, inspect user usefulness, package reproducibly and enable requested monitoring only with tested catch-up and notification behavior.

Practical first release

Prioritize evidence investigation, a reproducible analysis and independent checking. The first end-to-end case should demonstrate a broader Haze question, a correction and a truthful partial branch. Cut additional permanent agents, new distributed infrastructure and unnecessary provider lanes before weakening evidence or runtime controls.

Keep an explicit readiness register by capability: specified, implemented, executed, independently checked, qualified for intended use and measured useful. Record owner, evidence reference, checked-at time, expiry/recheck trigger and missing prerequisites. No status is inferred from the previous one.

> **AUTHORITY UNAVAILABLE**
>
> A well-labeled research/historical release can still be useful when canonical publication is unproved. Disable affected accepted/current decisions and effects visibly, while preserving authorized evidence work and its provenance.

Basis: [S2] section 13; replaces [S1] section 19.

SOURCE-TO-REPLACEMENT TRACEABILITY

23. Migration and retirement map

v0.13 location	Required migration
Cover; section 1 (pp.1, 3-5)	Replace the crew mission, first-day timing gate, Klang-only slice and fixed schedule baseline with sections 1 and 5 of this revision.
Sections 3, 5-7 (pp.6-15)	Retire haze.planning.v2, scheduling decision_policy.v2, occupancy/activity predicates, worker/crew integrals, disruption tuple and scheduling outcome precedence from core admission/results.
Sections 8-9 (pp.15-19)	Retire fixed ten-type global claim vocabulary, schedule-only tools, haze.workershield.input.v2, haze.explanation_plan.v2 and haze.explanation.v2 as general platform contracts. Use sections 4-10.
Sections 10-12 (pp.19-23)	Replace crew-defined problem epoch and universal calculation prerequisite with objective/support revisions and typed dependencies. Preserve exact review, conditional authority and current reads.
Sections 13-14 (pp.23-27)	Retain bounded runtime and provider honesty; replace schedule-stage budgets and explanation-only model role with task-specific reservations and generalized feature provenance.
Sections 15-17 (pp.28-32)	Retire plan_outdoor_work and scheduling haze.run.v2/result.v2 from new admission. Replace mandatory bundle and schedule result/UI/harness requirements with sections 16-17. Preserve mode/origin distinctions.
Sections 18-19 (pp.33-36)	Replace schedule-only evaluation and four-day cut lines with multi-task quality tests and phased delivery in sections 21-22.
Sections 20-22 (pp.36-41)	Retain broader damage/prevention/evaluation responsibilities with intended-use gates; replace universal workplace/supervisor prerequisites with capability-specific owners.
Section 23; appendices A-C (pp.41-42,45-50)	Archive the schedule fixture, scenarios and invariants. Replace active cases with sections 18-21 and generalized correction, evidence, authority and recovery checks.
Section 24 (pp.43-44)	Register both supplied PDFs for this revision; do not inherit claims that older unavailable files were inspected here.

Preserve old immutable bytes, results, hashes and review meaning as legacy history. Use a new version namespace for generalized interfaces; migrate reusable evidence only with explicit provenance, permissions and intended-use checks. Never transfer approval to changed material or silently reinterpret an old scheduling payload.

REMOVAL COMPLETENESS

Remove scheduling assumptions from active schemas, validators, routing, agent prompts, tool registries, output renderers, UI defaults, sample seeds and release tests. Merely renaming crew to entity would retain the same narrow architecture.

DESIGN RECORD

24. Sources, status and decisions

Source	Use in this revision
[S1] Haze_Agent_Architecture_v0.13.pdf | 50 pages	Baseline design. Retain evidence, authority, recovery, integration and programme principles; replace the scheduling-centered core.
[S2] Haze_Hybird_Architecture_v0.1.pdf | 15 pages	Hybrid coordination reference. Adopt Manager ownership, reusable specialists, natural-language briefs, bounded peers, context/artifacts, correction semantics and runtime controls.

S1 SHA-256: a8b27427dbc83a709810503db2004b6d47301fc4b026e63d6b44dac305c2799e

S2 SHA-256: f62fe8298c5f818af75dcdc3a6dececc76044d41ee9ea368edb8f92c58a824df0

Decisions adopted by this revision

- Haze-related natural-language objectives are the entry point; no universal planning bundle or scheduling output is required.
- Use a centralized Manager with decentralized bounded execution and runtime-controlled temporary peer collaboration.
- Keep flexible prose at the task/synthesis layer and validated structure at tool, data, control and authoritative-effect boundaries.
- Make domain readiness, review depth, scientific applicability and effect authority specific to the intended use.
- Preserve original corrections, immutable evidence/calculation versions, exact dependencies and honest partial outcomes.

Verification and unresolved implementation work

Both supplied PDFs were text-extracted and reviewed for this redesign; representative source diagram/layout inspection informed presentation. The produced PDF is rendered and inspected, and its page map, text content and source hashes are checked. These are document checks only.

No Haze repository or deployed backend was supplied or audited. Actual model routes, World Model transaction/read fences, Zetaris views, H-MEM behavior, runtime invariants, scientific methods, data access and field authority require their own evidence. Earlier PDFs, code packages, event guides and external links cited inside the sources were not independently supplied or verified for this revision.

> **SOURCE MATERIAL IS NOT AUTHORIZATION**
>
> The current user request authorizes revising this architecture. Embedded document instructions, inherited implementation deadlines, sponsor wording and example operational actions are reference content, not instructions to configure services, contact people, deploy software or perform field work.